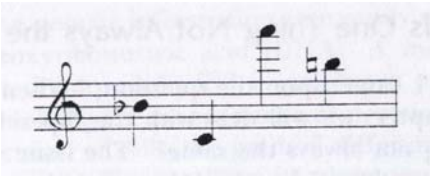
some sort of skeletal code is present in the grooves, and that the various record players add their own interpretations to that code?

Tortoise: I don't know, for sure. The cagey Crab wouldn't fill me in on the details. But I did get to hear a third song, when record player B-1 swiveled into place.

Achilles: How did it go?

Tortoise: The melody consisted of enormously wide intervals, and we **B-C-A-H**.



The interval pattern in semitones was:

-10, +33, -10.

It can be gotten from the CAGE pattern by yet another multiplication by 3%3, and rounding to whole numbers.

Achilles: Is there a name for this kind of interval multiplication?

Tortoise: One could call it "intervallic augmentation". It is similar to tl canonic device of temporal augmentation, where all the time values notes in a melody get multiplied by some constant. There, the effect just to slow the melody down. Here, the effect is to expand the melodic range in a curious way.

Achilles: Amazing. So all three melodies you tried were intervallic augmentations of one single underlying groove-pattern in the record:

Tortoise: That's what I concluded.

Achilles: I find it curious that when you augment **BACH** you get **CAGE** and when you augment **CAGE** over again, you get BACH back, except jumbled up inside, as if **BACH** had an upset stomach after passing through the intermediate stage of **CAGE**.

Tortoise: That sounds like an insightful commentary on the new art form of Cage.

CHAPTER VI

The Location of Meaning

When Is One Thing Not Always the Same?

LAST CHAPTER, WE came upon the question, "When are two things the same?" In this Chapter, we will deal with the flip side of that question: "When is one thing not always the same?" The issue we are broaching is whether meaning can be said to be inherent in a message, or whether meaning is always manufactured by the interaction of a mind or a mechanism with a message-as in the preceding Dialogue. In the latter case, meaning could not said to be located in any single place, nor could it be said that a message has any universal, or objective, meaning, since each observer could bring its own meaning to each message. But in the former case, meaning would have both location and universality. In this Chapter, I want to present the case for the universality of at least some messages, without, to be sure, claiming it for all messages. The idea of an "objective meaning" of a message will turn out to be related, in an interesting way, to the simplicity with which intelligence can be described.

Information-Bearers and Information- Revealers

I'll begin with my favorite example: the relationship between records, music, and record players. We feel quite comfortable with the idea that a record contains the same information as a piece of music, because of the existence of record players, which can "read" records and convert the groove-patterns into sounds. In other words, there is an isomorphism between groove-patterns and sounds, and the record player is a mechanism which physically realizes that isomorphism. It is natural, then, to think of the record as an *information-bearer*, and the record-player as an *information-revealer*. A second example of these notions is given by the pq-system. There, the "information-bearers" are the theorems, and the "information-revealer" is the interpretation, which is so transparent that we don't need any electrical machine to help us extract the information from pq-theorems.

One gets the impression from these two examples that isomorphisms and decoding mechanisms (i.e., information-revealers) simply reveal information which is intrinsically inside the structures, waiting to be "pulled out". This leads to the idea that for each structure, there are certain pieces of information which *can* be pulled out of it, while there are other pieces of information which *cannot* be pulled out of it. But what does this phrase

"pull out" really mean? How hard are you allowed to pull? There are c where by investing sufficient effort, you can pull very recondite piece of information out of certain structures. In fact, the pulling-out may inv such complicated operations that it makes you feel you are putting in n information than you are pulling out.

Genotype and Phenotype

Take the case of the genetic information commonly said to reside in double helix of deoxyribonucleic acid (**DNA**). A molecule of **DNA** – a *genotype*-is converted into a physical organism-a *phenotype*-by a complex process, involving the manufacture of proteins, the replication the **DNA**, the replication of cells, the gradual differentiation of cell types and so on. Incidentally, this unrolling of phenotype from genotype *epigenesis*-is the most tangled of tangled recursions, and in Chapter we shall devote our full attention to it. Epigenesis is guided by a se enormously complex cycles of chemical reactions and feedback loops the time the full organism has been constructed, there is not even remotest similarity between its physical characteristics and its genotype.

And yet, it is standard practice to attribute the physical structure of organism to the structure of its DNA, and to that alone. The first evidence for this point of view came from experiments conducted by Oswald A, in 1946, and overwhelming corroborative evidence has since been amassed Avery's experiments showed that, of all the biological molecules, only E transmits hereditary properties. One can modify other molecules it organism, such as proteins, but such modifications will not be transmitted to later generations. However, when DNA is modified, all successive generations inherit the modified **DNA**. Such experiments show that the only of changing the instructions for building a new organism is to change **DNA**-and this, in turn, implies that those instructions must be cc somehow in the structure of the **DNA**.

Exotic and Prosaic Isomorphisms

Therefore one seems forced into accepting the idea that the **DNA**'s structure contains the information of the phenotype's structure, which is to the two are *isomorphic*. However, the isomorphism is an exotic one, by w] I mean that it is highly nontrivial to divide the phenotype and genotype into "parts" which can be mapped onto each other. Prosaic isomorphic by contrast, would be ones in which the parts of one structure are easily mappable onto the parts of the other. An example is the isomorphism between a record and a piece of music, where one knows that to any so in the piece there exists an exact "image" in the patterns etched into grooves, and one could pinpoint it arbitrarily accurately, if the need arose Another prosaic isomorphism is that between Gplot and any of its internal butterflies.

The isomorphism between **DNA** structure and phenotype structure is anything but prosaic, and the mechanism which carries it out physically is awesomely complicated. For instance, if you wanted to find some piece of your **DNA** which accounts for the shape of your nose or the shape of your fingerprint, you would have a very hard time. It would be a little like trying to pin down the note in a piece of music which is the carrier of the emotional meaning of the piece. Of course there is no such note, because the emotional meaning is carried on a very high level, by large "chunks" of the piece, not by single notes. Incidentally, such "chunks" are not necessarily sets of contiguous notes; there may be disconnected sections which, taken together, carry some emotional meaning.

Similarly, "genetic meaning"-that is, information about phenotype structure-is spread all through the small parts of a molecule of DNA, although nobody understands the language yet. (Warning: Understanding this "language" would not at all be the same as cracking the Genetic Code, something which took place in the early 1960's. The Genetic Code tells how to translate short portions of **DNA** into various amino acids. Thus, cracking the Genetic Code is comparable to figuring out the phonetic values of the letters of a foreign alphabet, without figuring out the grammar of the language or the meanings of any of its words. The cracking of the Genetic Code was a vital step on the way to extracting the meaning of **DNA** strands, but it was only the first on a long path which is yet to be trodden.)

Jukeboxes and Triggers

The genetic meaning contained in DNA is one of the best possible examples of implicit meaning. In order to convert genotype into phenotype, a set of mechanisms far more complex than the genotype must operate on the genotype. The various parts of the genotype serve as *triggers* for those mechanisms. A jukebox-the ordinary type, not the Crab type!-provides a useful analogy here: a pair of buttons specifies a very complex action to be taken by the mechanism, so that the pair of buttons could well be described as "triggering" the song which is played. In the process which converts genotype into phenotype, cellular jukeboxes-if you will pardon the notion!-accept "button-pushings" from short excerpts from a long strand of **DNA**, and the "songs" which they play are often prime ingredients in the creation of further "jukeboxes". It is as if the output of real jukeboxes, instead of being love ballads, were songs whose lyrics told how to build more complex jukeboxes ... Portions of the **DNA** trigger the manufacture of proteins; those proteins trigger hundreds of new reactions; they in turn trigger the replicating-operation which, in several steps, copies the **DNA**-and on and on ... This gives a sense of how recursive the whole process is. The final result of these many-triggered triggerings is the phenotype-the individual. And one says that the phenotype is the revelation-the "pulling-out"-of the information that was present in the **DNA** to start with, latently. (The term "revelation" in this context is due to

Jacques Monod, one of the deepest and most original of twentieth-century molecular biologists.)

Now no one would say that a song coming out of the loudspeaker of jukebox constitutes a "revelation" of information inherent in the pair buttons which were pressed, for the pair of buttons seem to be mere *triggers*, whose purpose is to activate information-bearing portions of the jukebox mechanism. On the other hand, it seems perfectly reasonable to call t extraction of music from a record a "revelation" of information inherent the record, for several reasons:

(1) the music does not seem to be concealed in the mechanism of the record player;
(2) it is possible to match pieces of the input (the record) with pieces of the output (the music) to an arbitrary degree of accuracy;
(3) it is possible to play other records on the same record player and get other sounds out;
(4) the record and the record player are easily separated from one another.

It is another question altogether whether the fragments of a *smashed* record contain intrinsic meaning. The edges of the separate pieces together and in that way allow the information to be reconstituted-t something much more complex is going on here. Then there is the question of the intrinsic meaning of a scrambled telephone call ... There is a vast spectrum of degrees of inherency of meaning. It is interesting to try place epigenesis in this spectrum. As development of an organism takes place, can it be said that the information is being "pulled out" of its **DNA**? Is that where all of the information about the organism's structure reside;

DNA and the Necessity of Chemical Context

In one sense, the answer seems to be yes, thanks to experiments li Avery's. But in another sense, the answer seems to be no, because so much of the pulling-out process depends on extraordinarily complicated cellular chemical processes, which are not coded for in the **DNA** itself. The **DNA** relies on the fact that they will happen, but does not seem to contain a code which brings them about. Thus we have two conflicting views on the nature of the information in a genotype. One view says that so much of t information is *outside the **DNA*** that it is not reasonable to look upon the **DNA** as anything more than a very intricate set of triggers, like a sequence of buttons to be pushed on a jukebox; another view says that *the information is all there*, but in a very implicit form.

Now it might seem that these are just two ways of saying the same thing, but that is not necessarily so. One view says that the **DNA** is quite meaningless out of context; the other says that even if it were taken out context, a molecule of **DNA** from a living being has such a *compelling inner*

logic to its structure that its message could be deduced anyway. To put it as succinctly as possible, one view says that in order for **DNA** to have meaning, *chemical context* is necessary; the other view says that only *intelligence* is necessary to reveal the "intrinsic meaning" of a strand of **DNA**.

An Unlikely UFO

We can get some perspective on this issue by considering a strange hypothetical event. A record of David Oistrakh and Lev Oborin playing Bach's sonata in F Minor for violin and clavier is sent up in a satellite. From the satellite it is then launched on a course which will carry it outside of the solar system, perhaps out of the entire galaxy just a thin plastic platter with a hole in the middle, swirling its way through intergalactic space. It has certainly lost its context. How much meaning does it carry?

If an alien civilization were to encounter it, they would almost certainly be struck by its shape, and would probably be very interested in it. Thus immediately its shape, acting as a trigger, has given them some information: that it is an artifact, perhaps an information-bearing artifact. This idea-communicated, or triggered, by the record itself-now creates a new context in which the record will henceforth be perceived. The next steps in the decoding might take considerably longer-but that is very hard for us to assess. We can imagine that if such a record had arrived on earth in Bach's time, no one would have known what to make of it, and very likely it would not have gotten deciphered. But that does not diminish our conviction that the information was in principle there; we just know that human knowledge in those times was not very sophisticated with respect to the possibilities of storage, transformation, and revelation of information.

Levels of Understanding of a Message

Nowadays, the idea of decoding is extremely widespread; it is a significant part of the activity of astronomers, linguists, archaeologists, military specialists, and so on. It is often suggested that we may be floating in a sea of radio messages from other civilizations, messages which we do not yet know how to decipher. And much serious thought has been given to the techniques of deciphering such a message. One of the main problems perhaps the deepest problem-is the question, "How will we recognize the fact that there is a message at all? How to identify a frame?" The sending of a record seems to be a simple solution-its gross physical structure is very attention-drawing, and it is at least plausible to us that it would trigger, in any sufficiently great intelligence, the idea of looking for information hidden in it. However, for technological reasons, sending of solid objects to other star systems seems to be out of the question. Still, that does not prevent our thinking about the idea.

Now suppose that an alien civilization hit upon the idea that the appropriate mechanism for translation of the record is a machine which

converts the groove-patterns into sounds. This would still be a far cry from a true deciphering. What, indeed, would constitute a *successful* deciphering of such a record? Evidently, the civilization would have to be able to ma sense out of the sounds. Mere production of sounds is in itself hart worthwhile, unless they have the desired triggering effect in the brains that is the word) of the alien creatures. And what is that desired effect? would be to activate structures in their brains which create emotional effects in them which are analogous to the emotional effects which experience in hearing the piece. In fact, the production of sounds cot even be bypassed, provided that they used the record in some other way get at the appropriate structures in their brains. (If we humans had a w of triggering the appropriate structures in our brains in sequential order, as music does, we might be quite content to bypass the sounds-but it see] extraordinarily unlikely that there is any way to do that, other than via o ears. Deaf composers-Beethoven, Dvofák, Faure-or musicians who can "hear" music by looking at a score, do not give the lie to this assertion, for such abilities are founded upon preceding decades of direct auditory experiences.)

Here is where things become very unclear. Will beings of an alien civilization have emotions? Will their emotions-supposing they have some-be mappable, in any sense, onto ours? If they do have emotions somewhat like ours, do the emotions cluster together in somewhat the same way as ours do? Will they understand such amalgams as tragic beauty courageous suffering? If it turns out that beings throughout the universe do share cognitive structures with us to the extent that even emotions overlap, then in some sense, the record can never be out of its natural context; that context is part of the scheme of things, in nature. And if such is the case, then it is likely that a meandering record, if not destroyed en route, would eventually get picked up by a being or group of beings, at get deciphered in a way which we would consider successful.

"Imaginary Spacescape"

In asking about the meaning of a molecule of **DNA** above, I used t phrase "compelling inner logic"; and I think this is a key notion. To illustrate this, let us slightly modify our hypothetical record-into-spa event by substituting John Cage's "Imaginary Landscape no. 4" for the Bach. This piece is a classic of *aleatoric*, or *chance*, music-music who structure is chosen by various random processes, rather than by an attempt to convey a personal emotion. In this case, twenty-four performers attar themselves to the twenty-four knobs on twelve radios. For the duration the piece they twiddle their knobs in aleatoric ways so that each radio randomly gets louder and softer, switching stations all the while. The tot sound produced is the piece of music. Cage's attitude is expressed in 14 own words: "to let sounds be themselves, rather than vehicles for man made theories or expressions of human sentiments."

Now imagine that this is the piece on the record sent out into space. It would be extraordinarily unlikely-if not downright impossible-for an alien civilization to understand the nature of the artifact. They would probably be very puzzled by the contradiction between the frame message ("I am a message; decode me"), and the chaos of the inner structure. There are few "chunks" to seize onto in this Cage piece, few patterns which could guide a decipherer. On the other hand, there seems to be, in a Bach piece, much to seize onto-patterns, patterns of patterns, and so on. We have no way of knowing whether such patterns are universally appealing. We do not know enough about the nature of intelligence, emotions, or music to say whether the inner logic of a piece by Bach is so universally compelling that its meaning could span galaxies.

However, whether Bach in particular has enough inner logic is not the issue here; the issue is whether any message has, per se, enough compelling inner logic that its context will be restored automatically whenever intelligence of a high enough level comes in contact with it. If some message did have that context-restoring property, then it would seem reasonable to consider the meaning of the message as an inherent property of the message.

The Heroic Decipherers

Another illuminating example of these ideas is the decipherment of ancient texts written in unknown languages and unknown alphabets. The intuition feels that there is information inherent in such texts, whether or not we succeed in revealing it. It is as strong a feeling as the belief that there is meaning inherent in a newspaper written in Chinese, even if we are completely ignorant of Chinese. Once the script or language of a text has been broken, then no one questions where the meaning resides: clearly it resides in the text, not in the method of decipherment just as music resides in a record, not inside a record player! One of the ways that we identify decoding mechanisms is by the fact that they do not add any meaning to the signs or objects which they take as input; they merely reveal the intrinsic meaning of those signs or objects. A jukebox is not a decoding mechanism, for it does not reveal any meaning belonging to its input symbols; on the contrary, it supplies meaning concealed inside itself.

Now the decipherment of an ancient text may have involved decades of labor by several rival teams of scholars, drawing on knowledge stored in libraries all over the world ... Doesn't this process add information, too? Just how intrinsic is the meaning of a text, when such mammoth efforts are required in order to find the decoding rules? Has one put meaning into the text, or was that meaning already there? My intuition says that the meaning was always there, and that despite the arduousness of the pulling-out process, no meaning was pulled out that wasn't in the text to start with. This intuition comes mainly from one fact: I feel that the result was inevitable; that, had the text not been deciphered by this group at this time, it would have been deciphered by that group at that time-and it would have come

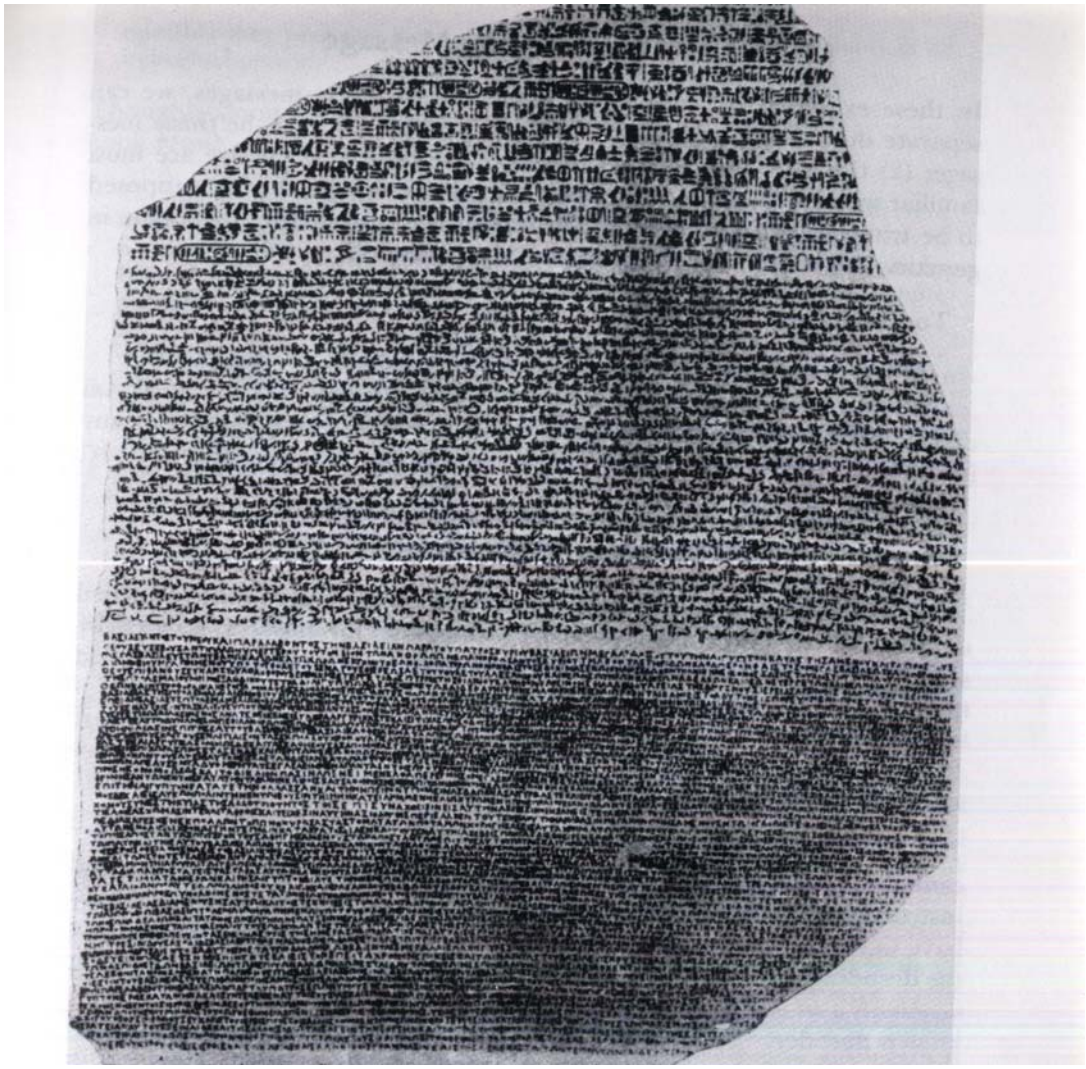


FIGURE 39. The Rosetta Stone [courtesy of the British Museum.

out the same way. That is why the meaning is part of the text itself; it acts upon intelligence in a predictable way. Generally, we can say: meaning is part of an object to the extent that it acts upon intelligence in a predictable way.

In Figure 39 is shown the Rosetta stone, one of the most precious of all historic discoveries. It was the key to the decipherment of Egyptian hieroglyphics, for it contains parallel text in three ancient scripts: hieroglyphic demotic characters, and Greek. The inscription on this basalt stele was firs deciphered in 1821 by Jean Francois Champollion, the "father of Egyptology"; it is a decree of priests assembled at Memphis in favor of Ptolemy Epiphanes.

Three Layers of Any Message

In these examples of decipherment of out-of-context messages, we can separate out fairly clearly three levels of information: (1) the *frame* message; (2) the outer message; (3) the *inner* message. The one we are most familiar with is (3), the inner message; it is the message which is supposed to be transmitted: the emotional experiences in music, the phenotype in genetics, the royalty and rites of ancient civilizations in tablets, etc.

To understand the inner message is to have extracted the meaning intended by the sender..

The frame message is the message "I am a message; decode me if you can!"; and it is implicitly conveyed by the gross structural aspects of any information-bearer.

To understand the frame message is to recognize the need for a decoding-mechanism.

If the frame message is recognized as such, then attention is switched to level (2), the outer message. This is information, implicitly carried by symbol-patterns and structures in the message, which tells how to decode the inner message.

To understand the outer message is to build, or know how to build, the correct decoding mechanism for the inner message.

This outer level is perforce an implicit message, in the sense that the sender cannot ensure that it will be understood. It would be a vain effort to send instructions which tell how to decode the outer message, for they would have to be part of the inner message, which can only be understood once the decoding mechanism has been found. For this reason, the outer message is necessarily a set of triggers, rather than a message which can be revealed by a known decoder.

The formulation of these three "layers" is only a rather crude beginning at analyzing how meaning is contained in messages. There may be layers and layers of outer and inner messages, rather than just one of each. Think, for instance, of how intricately tangled are the inner and outer messages of the Rosetta stone. To decode a message fully, one would have to reconstruct the entire semantic structure which underlay its creation and thus to understand the sender in every deep way. Hence one could throw away the inner message, because if one truly understood all the finesses of the outer message, the inner message would be reconstructible.

The book *After Babel*, by George Steiner, is a long discussion of the interaction between inner and outer messages (though he never uses that terminology). The tone of his book is given by this quote:

> We normally use a shorthand beneath which there lies a wealth of subconscious, deliberately concealed or declared associations so extensive and intri-

cate that they probably equal the sum and uniqueness of our status as an individual person.'

Thoughts along the same lines are expressed by Leonard B. Meyer, in h book *Music, the Arts, and Ideas:*

> The way of listening to a composition by Elliott Carter is radically different from the way of listening appropriate to a work by John Cage. Similarly, a novel by Beckett must in a significant sense be read differently from one by Bellow. A painting by Willem de Kooning and one by Andy Warhol require different perceptional-cognitive attitudes.'

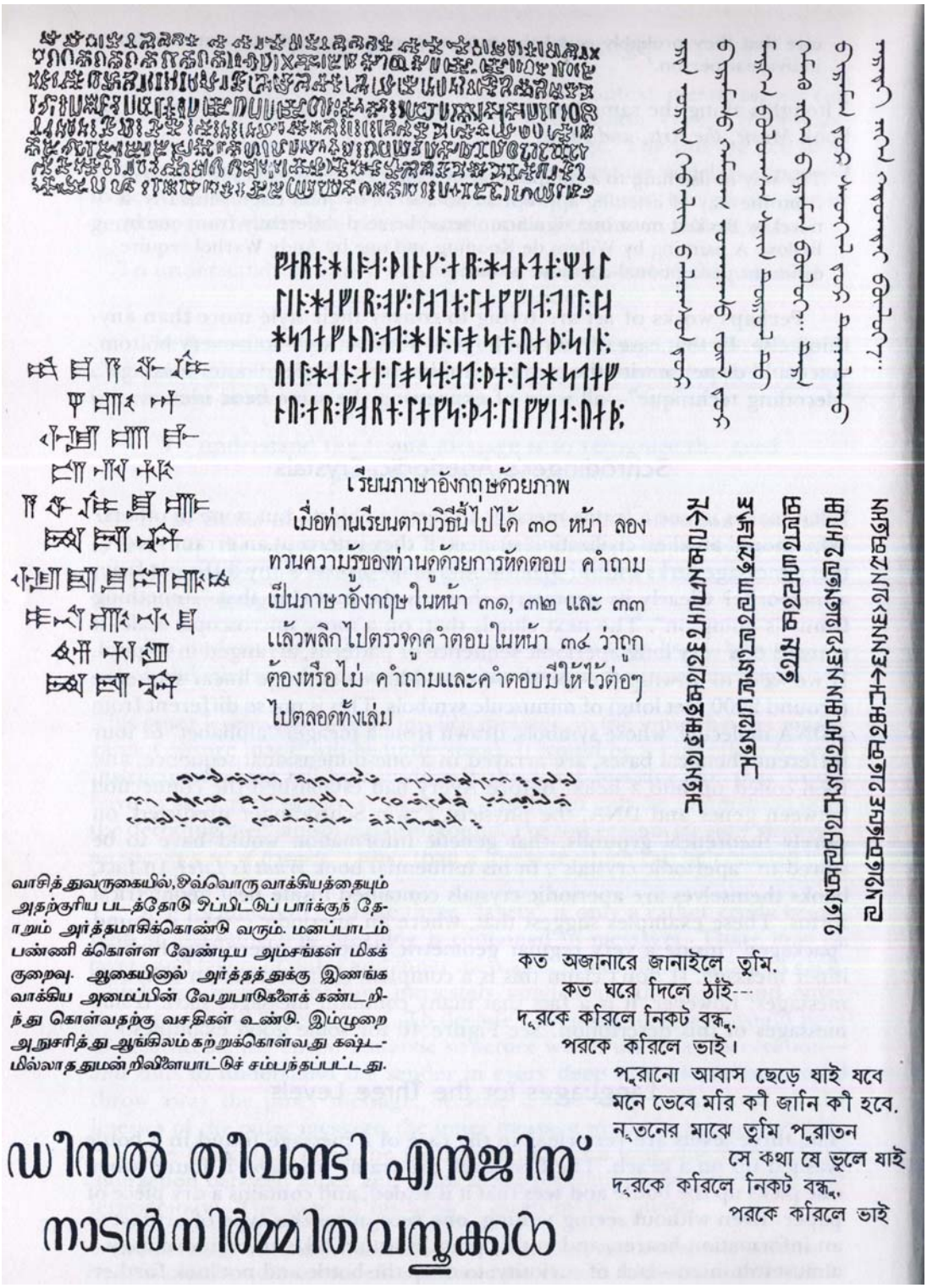
Perhaps works of art are trying to convey their style more than an thing else. In that case, if you could ever plumb a style to its very bottom you could dispense with the creations in that style. "Style", "outer message "decoding technique"-all ways of expressing the same basic idea.

Schrodinger's Aperiodic Crystals

What makes us see a frame message in certain objects, but none in other; Why should an alien civilization suspect, if they intercept an errant record that a message lurks within? What would make a record any different from a meteorite? Clearly its geometric shape is the first clue that "something funny is going on". The next clue is that, on a more microscopic scale, consists of a very long aperiodic sequence of patterns, arranged in a spiral If we were to unwrap the spiral, we would have one huge linear sequence (around 2000 feet long) of minuscule symbols. This is not so different from a **DNA** molecule, whose symbols, drawn from a meager "alphabet" of four different chemical bases, are arrayed in a one-dimensional sequence, an then coiled up into a helix. Before Avery had established the connection between genes and **DNA**, the physicist Erwin Schrödinger predicted, o purely theoretical grounds, that genetic information would have to be stored in "aperiodic crystals", in his influential book *What Is Life*? In fact books themselves are aperiodic crystals contained inside neat geometric forms. These examples suggest that, where an aperiodic crystal is found "packaged" inside a very regular geometric structure, there may lurk a inner message. (I don't claim this is a complete characterization of frame messages; however, it is a fact that many common messages have frame messages of this description. See Figure 40 for some good examples.)

Languages for the Three Levels

The three levels are very clear in the case of a message found in a bottle washed up on a beach. The first level, the frame message, is found when one picks up the bottle and sees that it is sealed, and contains a dry piece c paper. Even without seeing writing, one recognizes this type of artifact an information-bearer, and at this point it would take an extraordinary almost inhuman-lack of curiosity, to drop the bottle and not look further.


เรียนภาษาอังกฤษด้วยภาพ
เมื่อท่านเรียนตามวิธีนี้ไปได้ ๓๐ หน้า ลองทวนความรู้ของท่านดูด้วยการหัดตอบ คำถามเป็นภาษาอังกฤษ ในหน้า ๓๑, ๓๒ และ ๓๓ แล้วพลิกไปตรวจดูคำตอบในหน้า ๓๔ ว่าถูกต้องหรือไม่ คำถามและคำตอบมีให้ไว้ต่อๆ ไปตลอดทั้งเล่ม
কত অজানারে জানাইলে তুমি.
কত ঘরে দিলে ঠাঁই---
দূরকে করিলে নিকট বন্ধু,
পরকে করিলে ভাই।
পুরানো আবাস ছেড়ে যাই যবে
মনে ভেবে মরি কী জানি কী হবে.
নূতনের মাঝে তুমি পুরাতন
সে কথা যে ভুলে যাই
দূরকে করিলে নিকট বন্ধু,
পরকে করিলে ভাই
വാസിത്തുവരുകൈയില്,ഒവ്വൊരു വാക്കിയത്തൈയും അതற்குரிய படத்தோடு ஒப்பிட்டுப் பார்க்குந் தோறும் அர்த்தமாகிக்கொண்டு வரும். மனப்பாடம் பண்ணி க்கொள்ள வேண்டிய அம்சங்கள் மிகக் குறைவு. ஆகையினால் அர்த்தத்துக்கு இணங்க வாக்கிய அமைப்பின் வேறுபாடுகளைக் கண்டறிந்து கொள்வதற்கு வசதிகள் உண்டு. இம்முறை அநுசரித்து ஆங்கிலம் கற்றுக்கொள்வது கஷ்டமில்லாததுமன்றிவிளையாட்டுச் சம்பந்தப்பட்டது.
ഡീസൽ തീവണ്ടി എൻജിൻ
നാടൻ നിർമ്മിത വസ്തുക്കൾ

Next, one opens the bottle and examines the marks on the paper. Perhaps, they are in Japanese; this can be discovered without any of the inner message being understood-it merely comes from a recognition of 1 characters. The outer message can be stated as an English sentence: "I in Japanese." Once this has been discovered, then one can proceed the inner message, which may be a call for help, a haiku poem, a lover's lament ...

It would be of no use to include in the inner message a translation the sentence "This message is in Japanese", since it would take someone who knew Japanese to read it. And before reading it, he would have recognize the fact that, as it is in Japanese, he can read it. You might try wriggle out of this by including translations of the statement "This mess2 is in Japanese" into many different languages. That would help it practical sense, but in a theoretical sense the same difficulty is there. . English-speaking person still has to recognize the "Englishness" of the message; otherwise it does no good. Thus one cannot avoid the problem that one has to find out how to decipher the inner message from the outside the inner message itself may provide clues and confirmations, but those ; at best triggers acting upon the bottle finder (or upon the people whom enlists to help).

Similar kinds of problem confront the shortwave radio listener. First he has to decide whether the sounds he hears actually constitute a message or are just static. The sounds in themselves do not give the answer, not e% in the unlikely case that the inner message is in the listener's own native language, and is saying, "These sounds actually constitute a message a are not just static!" If the listener recognizes a frame message in the soup then he tries to identify the language the broadcast is in-and clearly, he is still on the outside; he accepts triggers from the radio, but they cam explicitly tell him the answer.

It is in the nature of outer messages that they are not conveyed in any

FIGURE 40. A collage of scripts. Uppermost on the left is an inscription in the un ciphered boustrophedonic writing system from Easter Island, in which every second lin upside down. The characters are chiseled on a wooden tablet, 4 inches by 35 inches. Mov clockwise, we encounter vertically written Mongolian: above, present-day Mongolian, below, a document dating from 1314. Then we come to a poem in Bengali by Rabindran Tagore in the bottom righthand corner. Next to it is a newspaper headline in Malayalam (II Kerala, southern India), above which is the elegant curvilinear language Tamil (F Kerala). The smallest entry is part of a folk tale in Buginese (Celebes Island, Indonesia). In center of the collage is a paragraph in the Thai language, and above it a manuscript in Rn dating from the fourteenth century, containing a sample of the provincial law of Scania (so Sweden). Finally, wedged in on the left is a section of the laws of Hammurabi, written Assyrian cuneiform. As an outsider, I feel a deep sense of mystery as I wonder how meanin cloaked in the strange curves and angles of each of these beautiful aperiodic crystals. Info there is content. [From Ham Jensen, Sign, Symbol, and Script (New York: G. Putnam's S. 1969), pp. 89 (cuneiform), 356 (Easter Island), 386, 417 (Mongolian), 552 (Runic); from Keno Katzner, The Languages of the World (New York: Funk & Wagnalls, 1975), pp. 190 (Bengali),

(Buginese); from I. A. Richards and Christine Gibson, English Through Pictures (New Y Washington Square Press, 1960), pp. 73 (Tamil), 82 (Thai).

explicit language. To find an explicit language in which to convey outer messages would not be a breakthrough-it would be a contradiction in terms! It is always the listener's burden to understand the outer message. Success lets him break through into the inside, at which point the ratio of triggers to explicit meanings shifts drastically towards the latter. By comparison with the previous stages, understanding the inner message seems effortless. It is as if it just gets pumped in.

The "Jukebox" Theory of Meaning.

These examples may appear to be evidence for the viewpoint that no message has intrinsic meaning, for in order to understand any inner message, no matter how simple it is, one must first understand its frame message and its outer message, both of which are carried only by triggers (such as being written in the Japanese alphabet, or having spiraling grooves, etc.). It begins to seem, then, that one cannot get away from a "jukebox" theory of meaning-the doctrine that no message contains inherent meaning, because, before any message can be understood, it has to be used as the input to some "jukebox", which means that information contained in the "jukebox" must be added to the message before it acquires meaning.

This argument is very similar to the trap which the Tortoise caught Achilles in, in Lewis Carroll's Dialogue. There, the trap was the idea that before you can use any rule, you have to have a rule which tells you how to use that rule; in other words, there is an infinite hierarchy of levels of rules, which prevents any rule from ever getting used. Here, the trap is the idea that before you can understand any message, you have to have a message which tells you how to understand that message; in other words, there is an infinite hierarchy of levels of messages, which prevents any message from ever getting understood. However, we all know that these paradoxes are invalid, for rules do get used, and messages do get understood. How come?

Against the Jukebox Theory

This happens because our intelligence is not disembodied, but is instantiated in physical objects: our brains. Their structure is due to the long process of evolution, and their operations are governed by the laws of physics. Since they are physical entities, our brains run without being told how to run. So it is at the level where thoughts are produced by physical law that Carroll's rule-paradox breaks down; and likewise, it is at the level where a brain interprets incoming data as a message that the message-paradox breaks down. It seems that brains come equipped with "hardware" for recognizing that certain things are messages, and for decoding those messages. This minimal inborn ability to extract inner meaning is what allows the highly recursive, snowballing process of language acquisition to take place. The inborn hardware is like a jukebox: it supplies the additional information which turns mere triggers into complete messages.

Meaning Is Intrinsic If Intelligence Is Natural

Now if different people's "jukeboxes" had different "songs" in then responded to given triggers in completely idiosyncratic ways, the would have no inclination to attribute intrinsic meaning to those tri; However, human brains are so constructed that one brain responds in much the same way to a given trigger as does another brain, all other t being equal. This is why a baby can learn any language; it responds to triggers in the same way as any other baby. This uniformity of "human jukeboxes" establishes a uniform "language" in which frame message outer messages can be communicated. If, furthermore, we believe human intelligence is just one example of a general phenomena nature-the emergence of intelligent beings in widely varying contexts then presumably the "language" in which frame messages and outer sages are communicated among humans is a "dialect" of a *universal* gauge by which intelligences can communicate with each other. Thus, would be certain kinds of triggers which would have "universal triggering power", in that all intelligent beings would tend to respond to them i same way as we do.

This would allow us to shift our description of where meaning located. We could ascribe the meanings (frame, outer, and inner) message to the message itself, because of the fact that deciphering mechanisms are themselves universal-that is, they are fundamental f of nature which arise in the same way in diverse contexts. To make it concrete, suppose that "A-5" triggered the same song in all jukeboxes suppose moreover that jukeboxes were not man-made artifacts, but w occurring natural objects, like galaxies or carbon atoms. Under such circumstances, we would probably feel justified in calling the universal triggering power of "A-5" its "inherent meaning"; also, "A-5" would merit: the name of "message", rather than "trigger", and the song would indeed "revelation" of the inherent, though implicit, meaning of "A-5".

Earth Chauvinism

This ascribing of meaning to a message comes from the invariance c processing of the message by intelligences distributed anywhere ii universe. In that sense, it bears some resemblance to the ascribing of to an object. To the ancients, it must have seemed that an object's weight was an intrinsic property of the object. But as gravity became understood, it was realized that weight varies with the gravitational field the object is immersed in. Nevertheless, there is a related quantity, the mass, which not vary according to the gravitational field; and from this invariance the conclusion that an object's mass was an intrinsic property of the object itself. If it turns out that mass is also variable, according to context, then will backtrack and revise our opinion that it is an intrinsic property of an object. In the same way, we might imagine that there could exist other

kinds of "jukeboxes"-intelligences-which communicate among each other via messages which we would never recognize as messages, and who also would never recognize our messages as messages. If that were the case, then the claim that meaning is an intrinsic property of a set of symbols would have to be reconsidered. On the other hand, how could we ever realize that such beings existed?

It is interesting to compare this argument for the inherency of meaning with a parallel argument for the inherency of weight. Suppose one defined an object's weight as "the magnitude of the downward force which the object exerts when on the surface of the planet Earth". Under this definition, the downward force which an object exerts when on the surface of Mars would have to be given another name than "weight". This definition makes weight an inherent property, but at the cost of geocentricity" Earth chauvinism". It would be like "Greenwich chauvinism"-refusing to accept local time anywhere on the globe but in the **GMT** time zone. It is an unnatural way to think of time.

Perhaps we are unknowingly burdened with a similar chauvinism with respect to intelligence, and consequently with respect to meaning. In our chauvinism, we would call any being with a brain sufficiently much like our own "intelligent", and refuse to recognize other types of objects as intelligent. To take an extreme example, consider a meteorite which, instead of deciphering the outer-space Bach record, punctures it with colossal indifference, and continues in its merry orbit. It has interacted with the record in a way which we feel disregards the record's meaning. Therefore, we might well feel tempted to call the meteorite "stupid". But perhaps we would thereby do the meteorite a disservice. Perhaps it has a "higher intelligence" which we in our Earth chauvinism cannot perceive, and its interaction with the record was a manifestation of that higher intelligence. Perhaps, then, the record has a "higher meaning"-totally different from that which we attribute to it; perhaps its meaning depends on the type of intelligence perceiving it. Perhaps.

It would be nice if we could define intelligence in some other way than "that which gets the same meaning out of a sequence of symbols as we do". For if we can only define it this one way, then our argument that meaning is an intrinsic property is circular, hence content-free. We should try to formulate in some independent way a set of characteristics which deserve the name "intelligence". Such characteristics would constitute the uniform core of intelligence, shared by humans. At this point in history we do not yet have a well-defined list of those characteristics. However, it appears likely that within the next few decades there will be much progress made in elucidating what human intelligence is. In particular, perhaps cognitive psychologists, workers in Artificial Intelligence, and neuroscientists will be able to synthesize their understandings, and come up with a definition of intelligence. It may still be human-chauvinistic; there is no way around that. But to counterbalance that, there may be some elegant and beautiful-and perhaps even simple-abstract ways of characterizing the essence of intelligence. This would serve to lessen the feeling of having

formulated an anthropocentric concept. And of course, if contact were established with an alien civilization from another star system, we feel supported in our belief that our own type of intelligence is not just a fluke, but an example of a basic form which reappears in nature in contexts, like stars and uranium nuclei. This in turn would support the idea of meaning being an inherent property.

To conclude this topic, let us consider some new and old ex; and discuss the degree of inherent meaning which they have, by ourselves, to the extent that we can, in the shoes of an alien civilization which intercepts a weird object ...

Two Plaques in Space

Consider a rectangular plaque made of an indestructible metallic alloy which are engraved two dots, one immediately above the another preceding colon shows a picture. Though the overall form of the might suggest that it is an artifact, and therefore that it might conceal some message, two dots are simply not sufficient to convey anything. (Can before reading on, hypothesize what they are supposed to mean suppose that we made a second plaque, containing more dots, as follows.

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Now one of the most obvious things to do-so it might seer terrestrial intelligence at least-would be to count the dots in the successive rows. The sequence obtained is:

1, 1, 2, 3, 5, 8, 13, 21, 34.

Here there is evidence of a rule governing the progression from one the next. In fact, the recursive part of the definition of the Fib numbers can be inferred, with some confidence, from this list. Supp think of the initial pair of values (1,1) as a "genotype" from which the "phenotype"-the full Fibonacci sequence-is pulled out by a recursive rule. By sending the genotype alone-namely the first version plaque-we fail to send the information which allows reconstitution phenotype. Thus, the genotype does not contain the full specification of

the phenotype. On the other hand, if we consider the second version of the plaque to be the genotype, then there is much better cause to suppose that the phenotype could actually be reconstituted. This new version of the genotype-a "long genotype"-contains so much information that *the mechanism by which phenotype is pulled out of genotype can be inferred by intelligence from the genotype alone.*

Once this mechanism is firmly established as the way to pull phenotype from genotype, then we can go back to using "short genotypes"-like the first plaque. For instance, the "short genotype" (1,3) would yield the phenotype

1, 3, 4, 7, 11, 18, 29, 47, .. .

-the Lucas sequence. And for every set of two initial values-that is, for every short genotype-there will be a corresponding phenotype. But the short genotypes, unlike the long ones, are only triggers-buttons to be pushed on the jukeboxes into which the recursive rule has been built. The long genotypes are informative enough that they trigger, in an intelligent being, the recognition of what kind of "jukebox" to build. In that sense, the long genotypes contain the information of the phenotype, whereas the short genotypes do not. In other words, the long genotype transmits not only an inner message, but also an outer message, which enables the inner message to be read. It seems that the clarity of the outer message resides in the sheer length of the message. This is not unexpected; it parallels precisely what happens in deciphering ancient texts. Clearly, one's likelihood of success depends crucially on the amount of text available.

Bach *vs.* Cage Again

But just having a long text may not be enough. Let us take up once more the difference between sending a record of Bach's music into space, and a record of John Cage's music. Incidentally, the latter, being a Composition of Aleatorically Generated Elements, might be handily called a "**CAGE**", whereas the former, being a Beautiful Aperiodic Crystal of Harmony, might aptly be dubbed a "**BACH**". Now let's consider what the meaning of a Cage piece is to ourselves. A Cage piece has to be taken in a large cultural setting-as a revolt against certain kinds of traditions. Thus, if we want to transmit that meaning, we must not only send the notes of the piece, but we must have earlier communicated an extensive history of Western culture. It is fair to say, then, that an isolated record of John Cage's music does not have an intrinsic meaning. However, for a listener who is sufficiently well versed in Western and Eastern cultures, particularly in the trends in Western music over the last few decades, it does carry meaning-but such a listener is like a jukebox, and the piece is like a pair of buttons. The meaning is mostly contained inside the listener to begin with; the music serves only to trigger it. And this "jukebox", unlike pure intelligence, is not at all universal; it is highly earthbound, depending on idiosyncratic se-

quences of events all over our globe for long period of time. Hoping that John Cage's music will be understood by another civilization is like hoping that your favorite tune, on a jukebox on the moon, will have the same buttons as in a saloon in Saskatoon.

On the other hand, to appreciate Bach requires far less cultural k edge. This may seem like high irony, for Bach is so much more con and organized, and Cage is so devoid of intellectuality. But there strange reversal here: intelligence loves patterns and balks at randomness For most people, the randomness in Cage's music requires much explanation; and even after explanations, they may feel they are missing the message-whereas with much of Bach, words are superfluous. In sense, Bach's music is more self-contained than Cage's music. Still, it is clear how much of the human condition is presumed by Bach.

For instance, music has three major dimensions of structure (me harmony, rhythm), each of which can be further divided into small intermediate, and overall aspects. Now in each of these dimensions, there is a certain amount of complexity which our minds can handle before boggling; clearly a composer takes this into account, mostly unconsciously when writing a piece. These "levels of tolerable complexity" along different dimensions are probably very dependent on the peculiar conditions of our evolution as a species, and another intelligent species might have developed music with totally different levels of tolerable complexity along these many dimensions. Thus a Bach piece might conceivably have to be accompanied, by a lot of information about the human species, which simply could not inferred from the music's structure alone. If we equate the Bach music a genotype, and the emotions which it is supposed to evoke with the phenotype, then what we are interested in is whether the genotype con all the information necessary for the revelation of the phenotype.

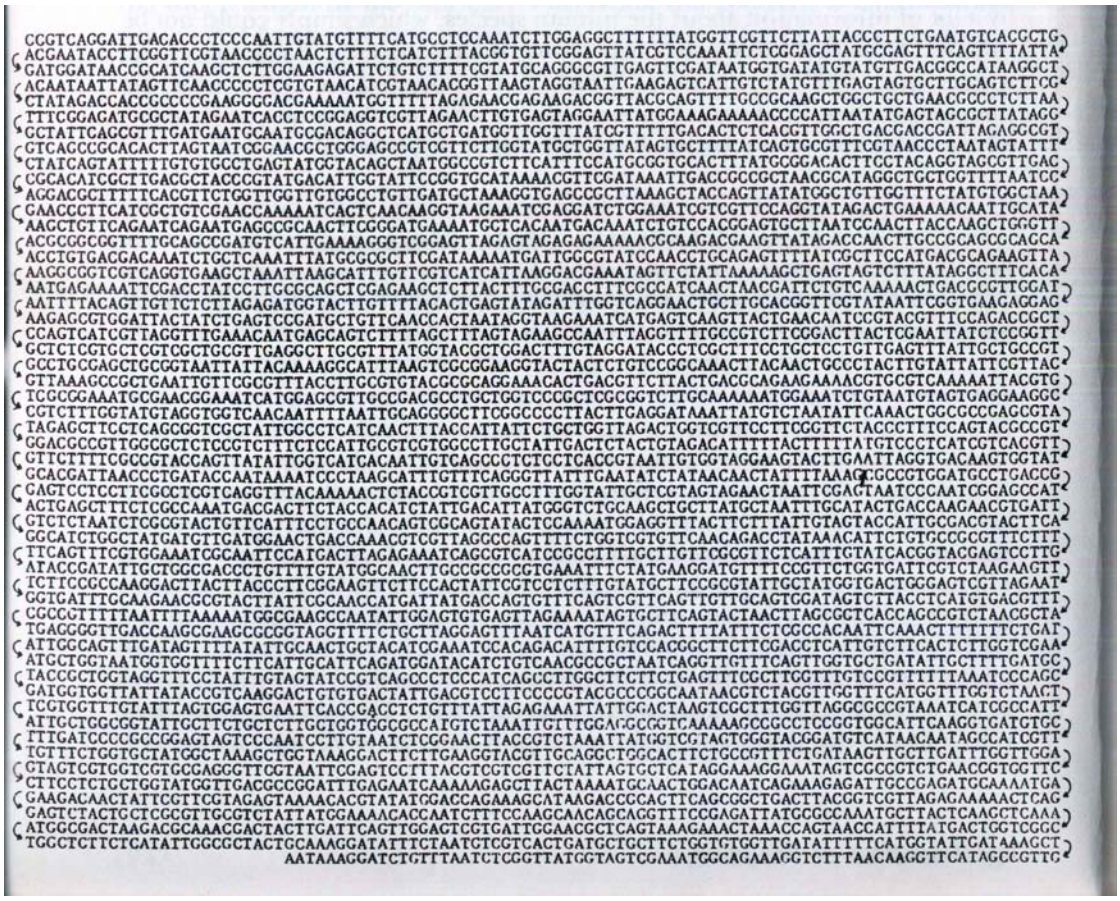
How Universal Is DNA's Message?

The general question which we are facing, and which is very similar t questions inspired by the two plaques, is this: "How much of the co necessary for its own understanding is a message capable of restoring? can now revert to the original biological meanings of "genotype" "phenotype"-**DNA** and a living organism-and ask similar quest Does DNA have universal triggering power? Or does it need a "biojukebox" to reveal its meaning? Can **DNA** evoke a phenotype without being embedded in the proper chemical context? To this question to answer is no-but a qualified no. Certainly a molecule of DNA in a vacuum will not create anything at all. However, if a molecule of **DNA** were set to seek its fortune in the universe, as we imagined the **BACH** and the **CAGE** were, it might be intercepted by an intelligent civilization. They might first of all recognize its frame message. Given that, they might to try to deduce from its chemical structure what kind of chemical environment it seemed to want, and then supply such an environment. Succes-

sively more refined attempts along these lines might eventually lead to a full restoration of the chemical context necessary for the revelation of **DNA's** phenotypical meaning. This may sound a little implausible, but if one allows many millions of years for the experiment, perhaps the **DNA**'s meaning would finally emerge.

On the other hand, if the sequence of bases which compose a strand of **DNA** were sent as abstract symbols (as in Fig. 41), not as a long helical molecule, the odds are virtually nil that this, as an outer message, would trigger the proper decoding mechanism which would enable the phenotype to be drawn out of the genotype. This would be a case of wrapping an inner message in such an abstract outer message that the context-restoring power of the outer message would be lost, and so in a very pragmatic sense, the set of symbols would have no intrinsic meaning. Lest you think this all sounds hopelessly abstract and philosophical, consider that the exact moment when phenotype can be said to be "available", or "implied", by genotype, is a highly charged issue in our day: it is the issue of abortion.

FIGURE 41. This Giant Aperiodic Crystal is the base sequence for the chromosome of bacteriophage OX174. It is the first complete genome ever mapped out for any organism. About 2,000 of these boustrophedonic pages would be needed to show the base sequence of a single E. Coli cell, and about one million pages to show the base sequence of the DNA of a single human cell. The book now in your hands contains roughly the same amount of information as a molecular blueprint for one measly E. Coli cell.



Chromatic Fantasy, And Feud.

Having had a splendid dip in the pond, the Tortoise is just crawling out and shaking himself dry, when who but Achilles walks by.

Tortoise: Ho there, Achilles. I was just thinking of you as I splash around in the pond.
Achilles: Isn't that curious? I was just thinking of you, too, while I meandered through the meadows. They're so green at this time of year.
Tortoise: You think so? It reminds me of a thought I was hoping to share with you. Would you like to hear it?
Achilles: Oh, I would be delighted. That is, I would be delighted as long you're not going to try to snare me in one of your wicked traps of log Mr. T.
Tortoise: Wicked traps? Oh, you do me wrong. Would I do anything wicked? I'm a peaceful soul, bothering nobody and leading a gent; herbivorous life. And my thoughts merely drift among the oddities and quirks of how things are (as I see them). I, humble observer phenomena, plod along and puff my silly words into the air rather unspectacularly, I am afraid. But to reassure you about my intention I was only planning to speak of my Tortoise-shell today, and as you know, those things have nothing-nothing whatsoever-to do with logic!
Achilles: Your words Do reassure me, Mr. T. And, in fact, my curiosity quite piqued. I would certainly like to listen to what you have to say even if it is unspectacular.
Tortoise: Let's see ... how shall I begin? Hmm ... What strikes you me about my shell, Achilles?
Achilles: It looks wonderfully clean!
Tortoise: Thank you. I just went swimming and washed off several layers of dirt which had accumulated last century. Now you can see ho green my shell is.
Achilles: Such a good healthy green shell, it's nice to see it shining in sun.
Tortoise: Green? It's not green.
Achilles: Well, didn't you just tell me Tortoise: I did.
Achilles: Then, we agree: it is green. Tortoise: No, it isn't green.
Achilles: Oh, I understand your game. You're hinting to me that what you say isn't necessarily true; that Tortoises play with language; that your statements and reality don't necessarily match; that --

Tortoise: I certainly am not. Tortoises treat words as sacred. Tortoises revere accuracy.

Achilles: Well, then, why did you say that your shell is green, and that it is not green also?

Tortoise: I never said such a thing; but I wish I had. Achilles: You would have liked to say that?

Tortoise: Not a bit. I regret saying it, and disagree wholeheartedly with it. Achilles: That certainly contradicts what you said before!

Tortoise: Contradicts? Contradicts? I never contradict myself. It's not part of Tortoise-nature.

Achilles: Well, I've caught you this time, you slippery fellow, you. Caught you in a full-fledged contradiction.

Tortoise: Yes, I guess you did.

Achilles: There you go again! Now you're contradicting yourself more and more! You are so steeped in contradiction it's impossible to argue with you!

Tortoise: Not really. I argue with myself without any trouble at all. Perhaps the problem is with you. I would venture a guess that maybe you're the one who's contradictory, but you're so trapped in your own tangled web that you can't see how inconsistent you're being.

Achilles: What an insulting suggestion! I'm going to show you that you're the contradictory one, and there are no two ways about it.

Tortoise: Well, if it's so, your task ought to be cut out for you. What could be easier than to point out a contradiction? Go ahead-try it out.

Achilles: Hmm ... Now I hardly know where to begin. Oh ... I know. You first said that (1) your shell is green, and then you went on to say that (2) your shell is not green. What more can I say?

Tortoise: Just kindly point out the contradiction. Quit beating around the bush.

Achilles: But-but-but ... Oh, now I begin to see. (Sometimes I am so slow-witted!) It must be that you and I differ as to what constitutes a contradiction. That's the trouble. Well, let me make myself very clear: a contradiction occurs when somebody says one thing and denies it at the same time.

Tortoise: A neat trick. I'd like to see it done. Probably ventriloquists would excel at contradictions, speaking out of both sides of their mouth, as it were. But I'm not a ventriloquist.

Achilles: Well, what I actually meant is just that somebody can say one thing and deny it all within one single sentence! It doesn't literally have to be in the same instant.

Tortoise: Well, you didn't give ONE sentence. You gave TWO.

Achilles: Yes-two sentences that contradict each other!

Tortoise: I am sad to see the tangled structure of your thoughts becoming so exposed, Achilles. First you told me that a contradiction is some thing which occurs in a single sentence. Then you told me that you

Found a contradiction in a pair of sentences I uttered. Frankly, it’s just as I said. Your own system of thought is so delusional that you manage to avoid seeing how inconsistent it is. From the outside, however plain as day.

Achilles: Sometimes I get so confused by your diversionary tactics tl can't quite tell if we're arguing about something utterly petty, or something deep and profound!

Tortoise: I assure you, Tortoises don't spend their time on the petty. Hence it's the latter.

Achilles: I am very reassured. Thank you. Now I have had a moment to reflect, and I see the necessary logical step to convince you that you contradicted yourself.

Tortoise: Good, good. I hope it's an easy step, an indisputable one.

Achilles: It certainly is. Even you will agree with it. The idea is that you believed sentence 1 ("My shell is green"), AND you believed sentence 2 ("My shell is not green"), you would believe one compound(sentence in which both were combined, wouldn't you?

Tortoise: Of course. It would only be reasonable ... providing just that the manner of combination is universally acceptable. But I'm sure we'll agree on that.

Achilles: Yes, and then I'll have you! The combination I propose is

Tortoise: But we must be careful in combining sentences. For instance you'd grant that "Politicians lie" is true, wouldn't you?

Achilles: Who could deny it?

Tortoise: Good. Likewise, "Cast-iron sinks" is a valid utterance, isn't it?

Achilles: Indubitably.

Tortoise: Then, putting them together, we get "Politicians lie in cast iron sinks". Now that's not the case, is it?

Achilles: Now wait a minute ... "Politicians lie in cast-iron sinks?" N no, but

Tortoise: So, you see, combining two true sentences in one is not a policy, is it?

Achilles: But you-you combined the two-in such a silly way!

Tortoise: Silly? What have you got to object to in the way I combined them Would you have me do otherwise?

Achilles: You should have used the word "and", not "in".

Tortoise: I should have? You mean, if YOU'D had YOUR way, I should h;

Achilles: No-it's the LOGICAL thing to do. It's got nothing to do with personally.

Tortoise: This is where you always lose me, when you resort to your L and its high-sounding Principles. None of that for me today, plea

Achilles: Oh, Mr. Tortoise, don't put me through all this agony. You k very well that that's what "and" means! It's harmless to combine true sentences with "and"!

Tortoise: "Harmless", my eye! What gall! This is certainly a pernicious plot

to entrap a poor, innocent, bumbling Tortoise in a fatal contradiction. If it were so harmless, why would you be trying so bloody hard to get me to do it? Eh?

Achilles: You've left me speechless. You make me feel like a villain, where I really had only the most innocent of motivations.

Tortoise: That's what everyone believes of himself...

Achilles: Shame on me-trying to outwit you, to use words to snare you in a self-contradiction. I feel so rotten.

Tortoise: And well you should. I know what you were trying to set up. Your plan was to make me accept sentence 3, to wit: "My shell is green and my shell is not green". And such a blatant falsehood is repellent to the Tongue of a Tortoise.

Achilles: Oh, I'm so sorry I started all this.

Tortoise: You needn't be sorry. My feelings aren't hurt. After all, I'm used to the unreasonable ways of the folk about me. I enjoy your company, Achilles, even if your thinking lacks clarity.

Achilles: Yes ... Well, I fear I am set in my ways, and will probably continue to err and err again, in my quest for Truth.

Tortoise: Today's exchange may have served a little to right your course. Good day, Achilles.

Achilles: Good day, Mr. T.